Binary Trees

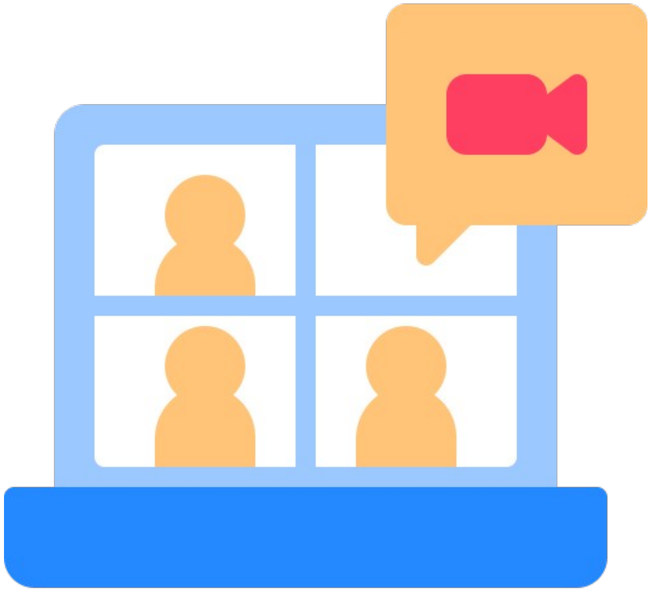
TIP101

Agenda

1	Welcome 👋	10 minutes
2	Binary Trees	35 minutes
3	**BREAK**	5 minutes
4	Collaborative Problem Solving	60 minutes
5	Wrap Up	10 minutes

10 minutes

Welcome! 👋



Turn on your cameras

Fill out the entry ticket

Icebreaker

2 minutes

📱**Question:**

Have you ever tried or wanted to try mobile development? Are you team iPhone or Android?

💬**Answer:**

Share your answer in the Zoom chat below. What kinds of pets do you have if any? If you don't have any, share what type you would like to have, if any.

Your Feedback

Glows

+ Recursion foundations went well, building on that this week!
+ Students appreciated peer to peer experience

Grows

- Interview + Breakout rooms seems a bit heavy

- [Followup] More Interview Prep Resources: **Hackerank**

Announcements & Reminders

- Problem Set Validator email sent out a few hours ago
  - We will demo right before the break!
  - Totally optional tool that we built based on you guys’ feedback.

35 minutes

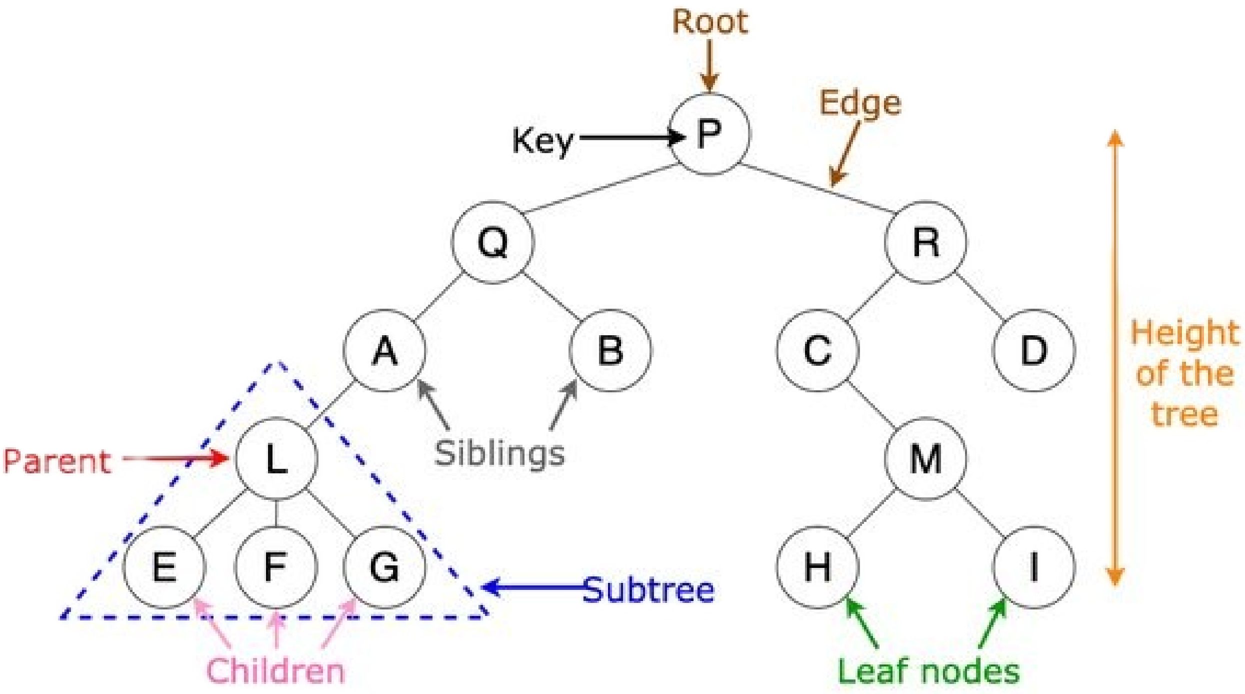
Binary Trees

Discussion

In the chat . . .

What do you already know about binary trees or binary search trees?

What is a Binary Tree?



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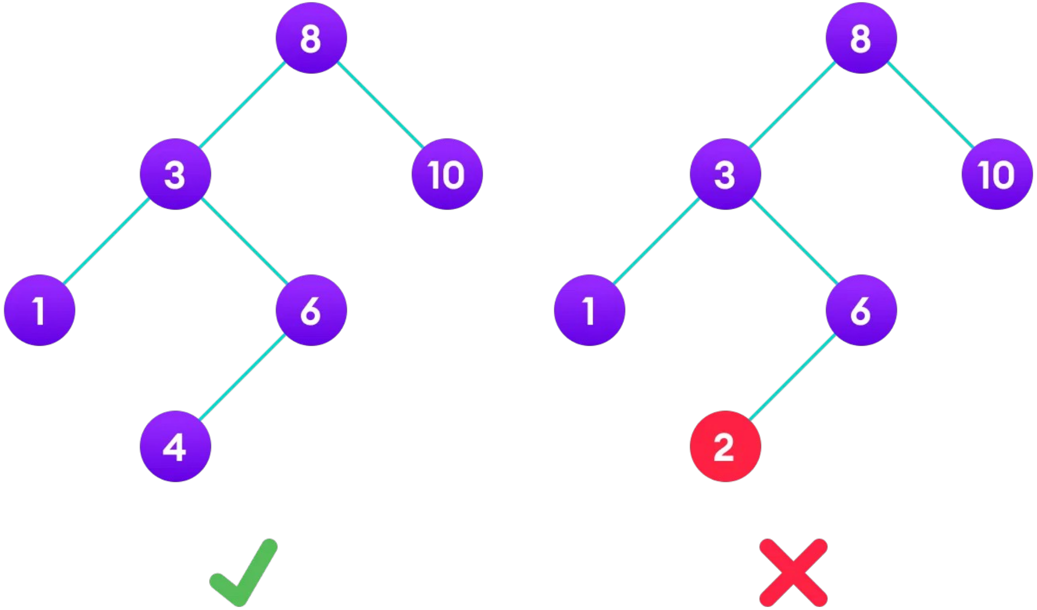
Discussion

In the chat . . .

What are some real-world examples where a binary tree structure might be useful?

Binary Search Trees (BSTs)



Check for Understanding

How does the ordering in a BST help with search operations?

A. It allows you to search only the left subtree for all values.

B. It ensures that all values are stored in a sorted array.

C. It allows you to decide whether to go left or right based on the value you're searching for, reducing the number of nodes to check.

D. It makes the tree balanced automatically, leading to faster search times.

Check for Understanding

Why might you choose a BST over a general binary tree?

A. A BST requires less memory than a general binary tree.

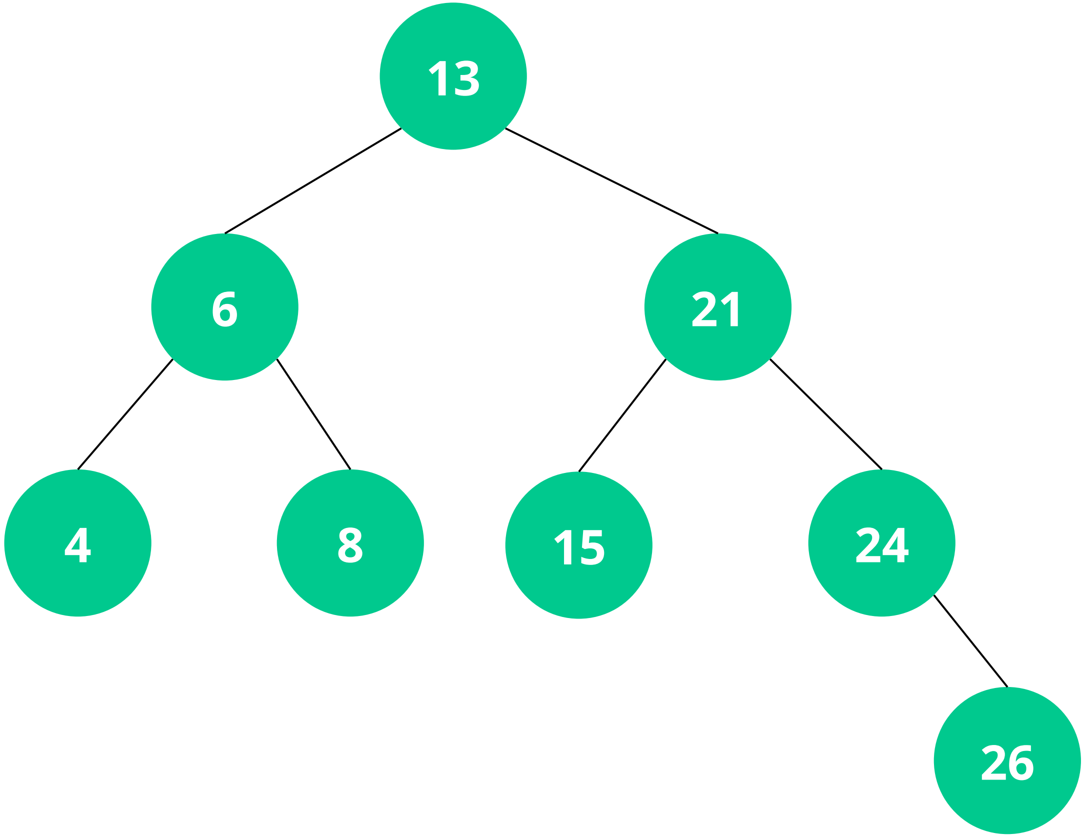
B. A BST maintains an ordered structure, allowing for efficient search, insertion, and deletion operations.

C. A BST automatically balances itself, leading to faster operations.

D. A BST can store more nodes than a general binary tree.

The TreeNode Class

Each **`TreeNode`** contains a value, and references to its left and right children.



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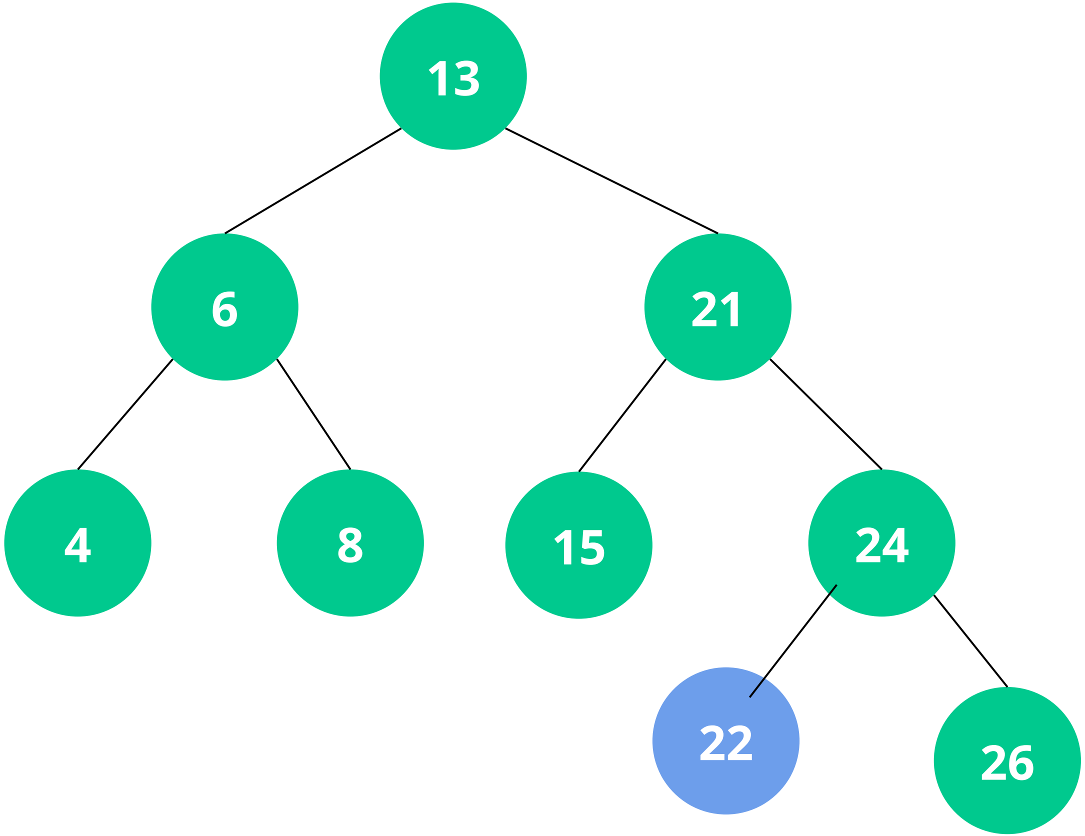
Node Insertion

Value to Insert: 22



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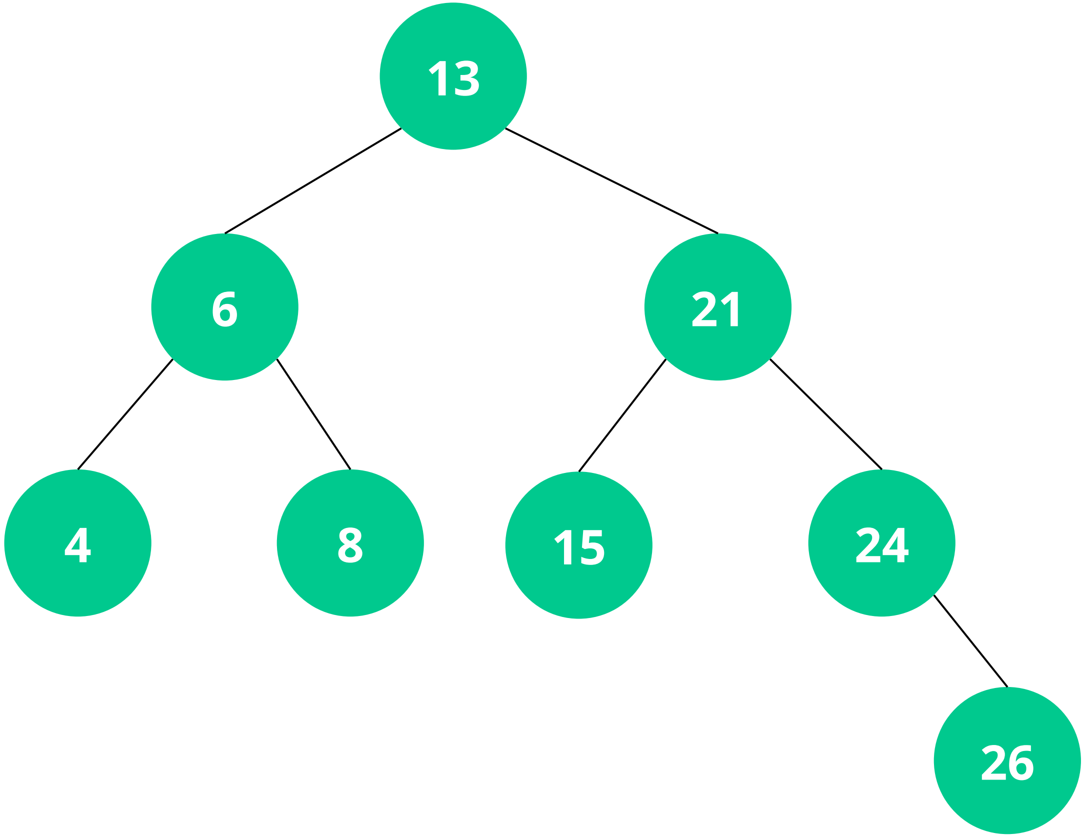
Node Insertion Algorithm: Pseudocode

1. Start at the root.
2. Compare the value to be inserted with the current node’s value.
3. If the value is less, move to the left child; if greater, move to the right child.
4. Repeat steps 2-3 until the correct position is found.
5. Insert the new node as a leaf.

Node Deletion

Value to Delete: 15



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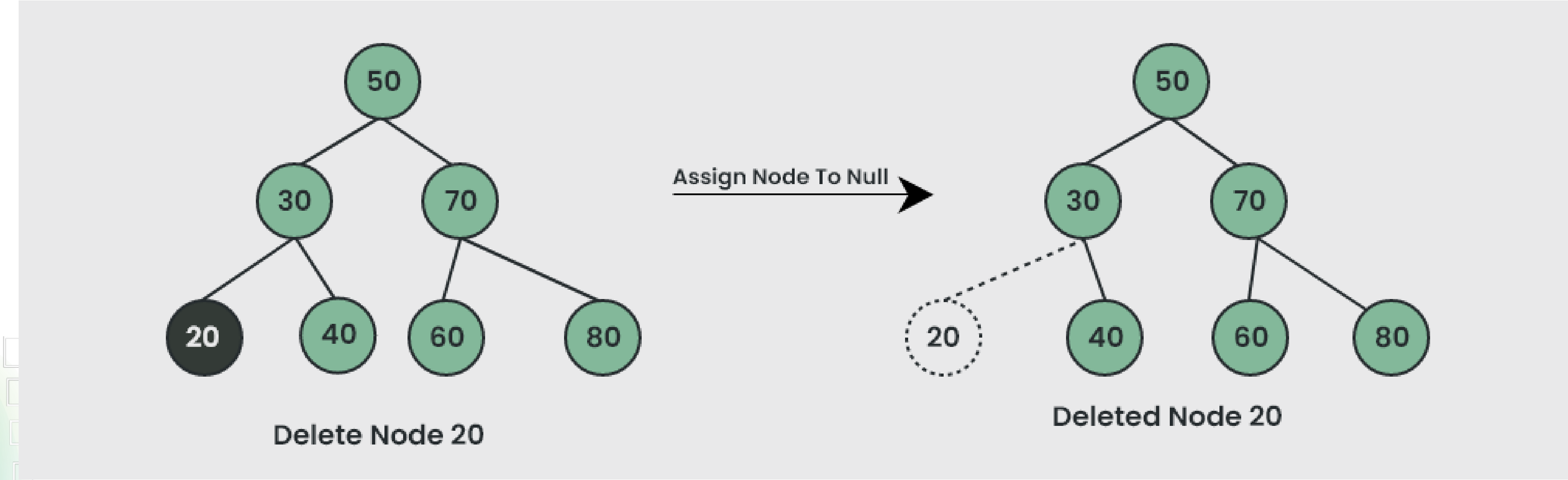
Deletion Cases

Three main cases to consider when deleting a node:

* Node has no children (leaf node).
* Node has one child.
* Node has two children.

Case 1: Node with No Children



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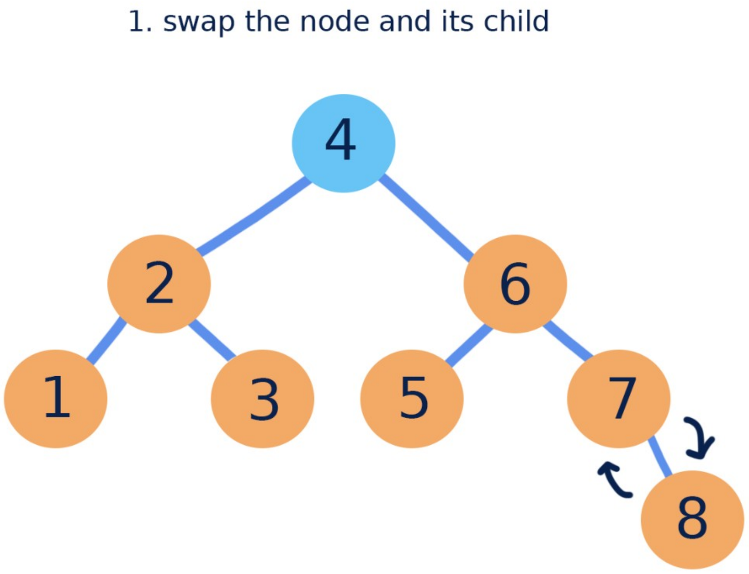
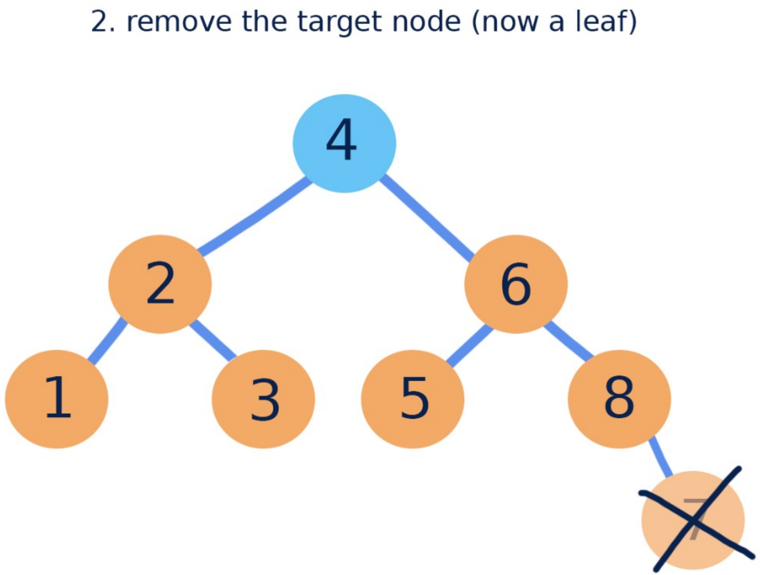
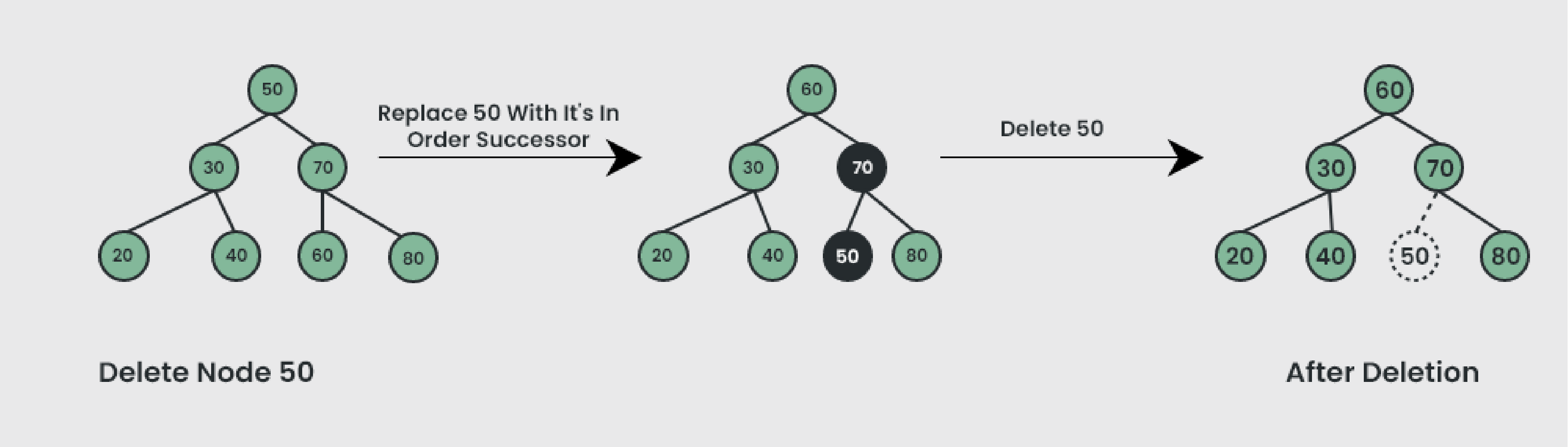
Case 2: Node with One Child





Case 3: Node with Two Children



Check for Understanding

Why is finding the minimum value node in the right subtree important when deleting a node with two children?

A. It ensures that the tree remains balanced after deletion.

B. It helps in finding the largest node in the right subtree.

C. It replaces the deleted node with the smallest value greater than the deleted node, maintaining the BST properties.

D. It simplifies the deletion process by removing a leaf node.

Problem Set Validator Demo

- Problem Set Validator email sent out a few hours ago
  - We will demo right before the break!
  - Totally optional tool that we built based on you guys' feedback.

Break Time!

Take 5 mins to step away from the computer. Feel free to turn off your camera and return promptly after 5 mins.

60 minutes

Collaborative Problem Solving

Breakout Rooms

60 minutes

1. 🏃 Navigate to your breakout room.
2. 👋 Introduce yourselves!
3. 👦💻 Work on the problems for your selected unit.

10 minutes

Wrap Up

One Good Thing

Let's reflect on today's session. Maybe you got a question answered, overcame a specific coding challenge, or learned how to do something completely new!

In the chat, share at least one good thing you've learned or experienced during the session today.

Let's celebrate together!

Session Survey Time!

Final Reminders

- [ ] Finish the problem sets for additional practice
- [ ] Take the weekly HackerRank assessment by Sunday @ midnight
- [ ] Visit a **Study Hall Session** to deepen your understanding, work through challenges, and connect with course material alongside your peers!